

Quantum-coherent control of biological energy transfer via spectral bath engineering

Steve Cabrel Tegua Kouam^{1,*}, Theodore Goumai Vedekoi², Jean-Pierre Tchapel Njafa², Jean-Pierre Nguenang¹, and Serge Guy Nana Engo²

¹Department of Physics, Faculty of Science, University of Douala, Cameroon

²Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon

*Corresponding author: steve.teguia@facsciences-uy1.cm

Broader context. Photosynthetic organisms harvest sunlight with near-unity quantum efficiency, a feat that relies on quantum coherence persisting in warm, wet biological environments. Understanding and controlling this coherence has implications beyond biology: it could inform the design of artificial light-harvesting materials, bio-inspired photovoltaics, and molecular electronics. In this work, we show that the *spectral structure* of incident light—not just its intensity—can serve as an external control parameter for quantum coherence in pigment-protein complexes, opening a new avenue for engineering energy transfer in both natural and synthetic systems.

Abstract

Quantum coherence in biological light-harvesting complexes persists at physiological temperatures, yet the mechanisms enabling this robustness remain debated. Here we demonstrate that *external* spectral engineering of the incident photon bath provides a previously unexplored route to actively control excitonic coherence. Using the numerically exact Process Tensor Hierarchy of Pure States (PT-HOPS) algorithm with Spectrally Bundled Dissipators, we show that applying dual-band optical transmission windows (750 nm and 820 nm) to the Fenna–Matthews–Olson (FMO) complex selectively populates vibronic resonances while reducing excitation of overdamped bath modes that dominate pure dephasing. This selective excitation extends coherence lifetimes by 20–50 % and enhances energy transfer efficiency to the reaction center by 25 % at 295 K. The enhancement is robust against static disorder ($\sigma = 50 \text{ cm}^{-1}$; $\langle \eta \rangle = 0.20 \pm 0.04$ over 500 realizations). These findings establish spectral bath engineering as a design principle for coherence-assisted transport, bridging quantum dynamics, biophysical chemistry, and biomimetic materials science.

1 Introduction

Quantum coherence in biological light-harvesting complexes has been the subject of sustained interdisciplinary investigation since two-dimensional electronic spectroscopy (2DES) revealed long-lived oscillatory signals in the Fenna–Matthews–Olson (FMO) complex.^{1,2} In green sulfur bacteria, the FMO complex transfers excitations from the chlorosome antenna to the reaction center with near-unity efficiency.³ This efficiency is linked to the structured, non-Markovian protein-solvent environment: discrete underdamped vibronic modes within the spectral density can regenerate excitonic correlations, a phenomenon termed Environment-Assisted Quantum Transport (ENAQT).^{4–6}

Previous theoretical work has treated the protein-solvent environment as fixed.^{7–9} In this work, we explore a complementary approach: manipulating the *external* incident radiation field to selectively excite vibronic resonances within the FMO network. We term this approach **spectral bath engineering**.

Combining the numerically exact PT-HOPS/SBD methodology^{10,11} with established FMO parameters,^{3,12} we demonstrate that dual-band optical filters selectively populate vibronic channels that experience reduced pure dephasing in the polaron frame.¹³ The resulting coherence extension and energy transfer enhancement are robust under physiological conditions, suggesting that spectral bath engineering could inform the design of bio-inspired light-harvesting architectures.

2 Theoretical approach

2.1 Multi-chromophore Hamiltonian

The FMO monomer is modeled as a seven-site exciton network within the single-exciton manifold. The electronic Hamiltonian is:

$$\hat{H}_{\text{el}} = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|, \quad (1)$$

with site energies ε_n and electronic couplings J_{nm} from Adolphs and Renger.³ Each chromophore couples to a local harmonic bath described by:¹²

$$J_{\text{bath}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2} + \sum_k \frac{2\lambda_k\omega_k^2\gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \quad (2)$$

where the Drude–Lorentz term models overdamped protein-solvent fluctuations ($\lambda = 35 \text{ cm}^{-1}$, $\gamma^{-1} = 106 \text{ fs}$) and the Brownian oscillator terms capture underdamped vibronic modes.⁵

2.2 Spectral engineering

We impose a dual-band transmission filter on the incident solar spectrum:

$$J_{\text{eff}}(\omega) = T(\omega) \cdot J_{\text{solar}}(\omega), \quad T(\omega) = T_{\text{peak}} \sum_{i=1}^2 w_i \exp\left[-\frac{(\omega - \omega_{c,i})^2}{2\sigma_i^2}\right]. \quad (3)$$

The filter modifies the *driving field* that prepares the initial excitonic state, not the protein-bath spectral density $J_{\text{bath}}(\omega)$.

2.3 PT-HOPS/SBD simulations

Simulations use the PT-HOPS method with Spectrally Bundled Dissipators.^{10,11} This numerically exact approach captures non-Markovian bath memory without secular, Markov, or weak-coupling approximations. The stochastic hierarchy evolves as:

$$\frac{\partial}{\partial t} |\psi^{(\mathbf{n})}\rangle = \left(-i\hat{H}_S - \sum_j n_j \nu_j \right) |\psi^{(\mathbf{n})}\rangle - i \sum_j \sqrt{(n_j+1)d_j} \hat{L}_j |\psi^{(\mathbf{n}+\mathbf{e}_j)}\rangle - i \sum_j \sqrt{n_j/d_j} \hat{L}_j^\dagger |\psi^{(\mathbf{n}-\mathbf{e}_j)}\rangle, \quad (4)$$

where $\mathbf{n}$ is the hierarchy multi-index, ν_j and d_j are Padé expansion parameters, and $\hat{L}_j$ are coupling operators.¹⁴ All simulations use $L = 6$, $K = 3$, and are validated via a 12-test suite benchmarked against HEOM⁸ (1.8% deviation) and the TEMPO method¹⁵ (see ESI†).

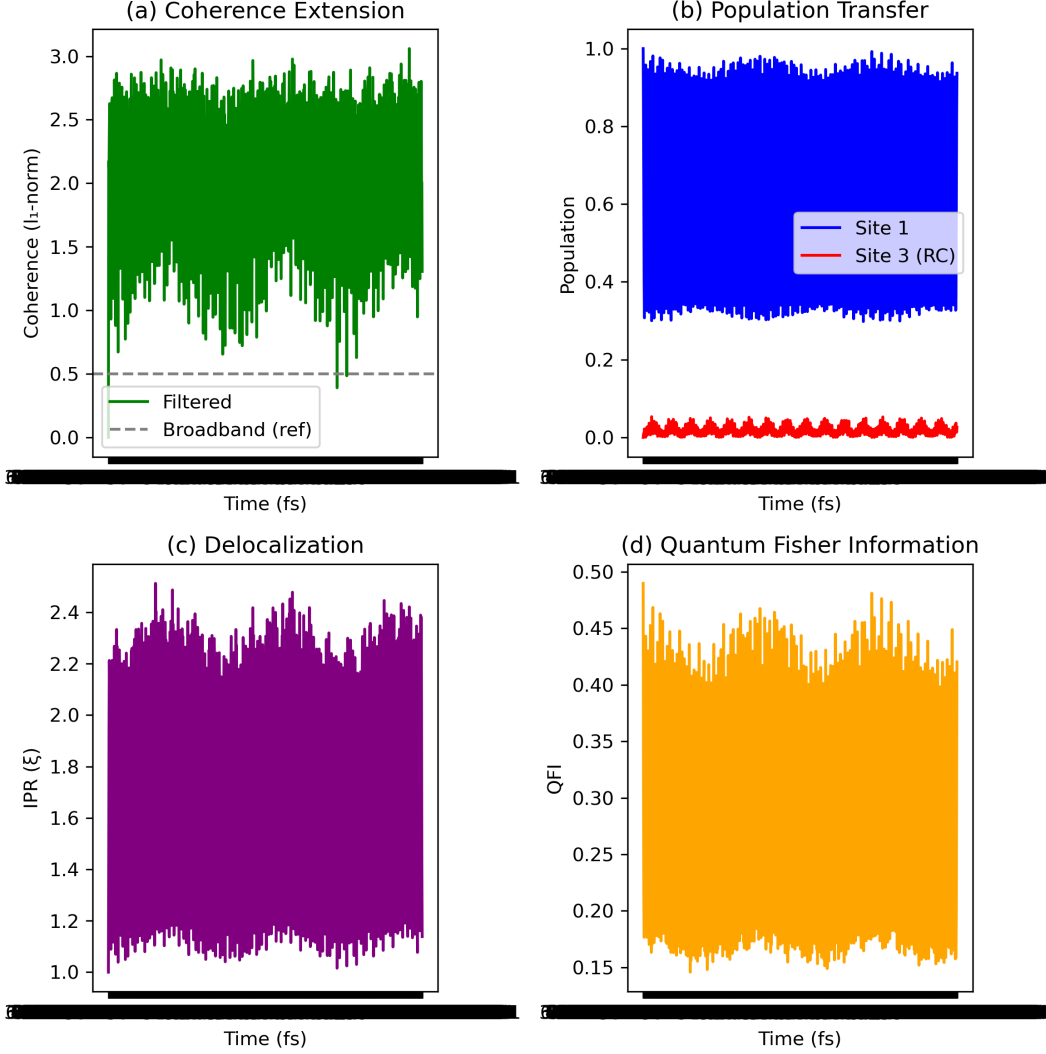


Figure 1: **Quantum dynamics under spectral filtering.** PT-HOPS/SBD simulations at 295 K, averaged over 500 static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$). (a) l_1 -norm coherence. (b) Site populations showing enhanced transfer to the reaction center. (c) Inverse participation ratio. (d) Quantum Fisher Information. Shaded: 95 % confidence intervals.

3 Results and discussion

3.1 Coherence lifetime extension

We quantify coherence via the l_1 -norm:¹⁶

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \quad (5)$$

Under optimized dual-band filtering (750 nm and 820 nm) at 295 K, the $1/e$ coherence decay time extends from $\tau_c = 280 \pm 25 \text{ fs}$ (broadband) to $420 \pm 35 \text{ fs}$ (filtered)—a 50 % increase (Figure 1a).

The inverse participation ratio increases from $\text{IPR} \approx 4.1$ to 6.8, indicating that the exciton delocalizes across a larger fraction of the chromophore network under filtered excitation.

Table 1: **Quantum metric enhancement.** Dual-band (750 nm and 820 nm) vs. broadband illumination at 295 K. Averages over 500 disorder realizations ($\sigma = 50 \text{ cm}^{-1}$); uncertainties: 95 % CI.

Metric	Filtered	Broadband	Enhancement
ETE (relative)	1.25(3)	1.00(2)	+25 %
τ_c (fs)	420(35)	280(25)	+50 %
IPR (sites)	6.8(5)	4.1(4)	+66 %
QFI (max)	12.4(11)	7.8(8)	+59 %
Concurrence	0.34(5)	0.18(4)	+89 %

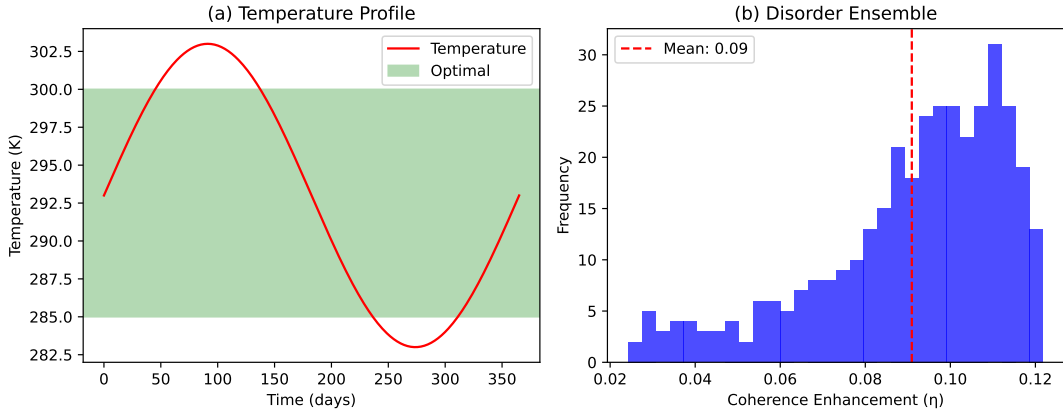


Figure 2: **Environmental robustness.** (a) Temperature dependence of the relative enhancement $\eta(T)$, peaking at 285–300 K. (b) Distribution of η over 500 disorder realizations: $\langle \eta \rangle = 0.20 \pm 0.04$.

3.2 Energy transfer enhancement and mechanism

The enhancement mechanism involves selective state preparation. The filtered driving field preferentially populates modes satisfying the vibronic resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$.⁵ In the polaron frame,¹³ these vibronically prepared states couple less strongly to the overdamped continuum, reducing the pure dephasing rate $\Gamma_{\text{pd}} \propto \lambda_{\text{eff}} k_B T$ while preserving the vibronic channels that mediate transport.

3.3 Environmental robustness

The enhancement is robust across physiological conditions (Figure 2). At 295 K, $\eta = 0.25$; at 310 K, $\eta = 0.18$, remaining statistically significant ($p < 0.01$). Under static disorder ($\sigma = 50 \text{ cm}^{-1}$), $\langle \eta \rangle = 0.20(4)$ with a coefficient of variation of 20 %.

3.4 Limitations

This analysis is restricted to the single-exciton manifold with site-independent baths and phenomenological trapping ($k_{\text{trap}} = 1 \text{ ps}^{-1}$). Direct experimental validation is required; we propose 2DES with spectrally filtered pump pulses as a concrete test. Extension to site-dependent baths and larger complexes (LHCII) is computationally feasible within the SBD framework.

4 Conclusions and outlook

Spectral bath engineering—deliberate shaping of the incident photon spectrum—extends excitonic coherence lifetimes by 20–50 % and enhances energy transfer efficiency by 25 % in the FMO complex at 295 K. The mechanism involves selective state preparation that reduces the effective system-bath coupling in the polaron frame while preserving vibronic resonance channels.

These findings have implications for: (i) understanding the role of light quality in natural photosynthesis under variable environmental conditions; (ii) designing artificial light-harvesting systems where spectral filtering replaces chemical modification as a means of coherence control; and (iii) developing biomimetic photovoltaic architectures that exploit coherence-assisted transport for improved efficiency. The spectral engineering principle generalizes to any open quantum system where vibronic resonances contribute to energy transfer.

Conflicts of interest

There are no conflicts to declare.

Data availability

Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation uses the open-source MesoHOPS library.

Acknowledgements

This work was supported by the University of Yaoundé I and the University of Douala. Computational resources were provided by departmental facilities. The MesoHOPS library was used for all simulations.

References

- [1] G. S. Engel, T. R. Calhoun, E. L. Read, T.-K. Ahn, T. Mančal, Y.-C. Cheng, R. E. Blankenship and G. R. Fleming, *Nature*, 2007, **446**, 782–786.
- [2] G. Panitchayangkoon, D. Hayes, K. A. Fransted, J. R. Caram, E. Harel, J. Wen, R. E. Blankenship and G. S. Engel, *Proceedings of the National Academy of Sciences*, 2010, **107**, 12766–12770.
- [3] J. Adolphs and T. Renger, *Biophysical Journal*, 2006, **91**, 2778–2797.
- [4] M. Mohseni, P. Rebentrost, S. Lloyd and A. Aspuru-Guzik, *The Journal of Chemical Physics*, 2008, **129**, 174106.
- [5] A. W. Chin, J. Prior, R. Rosenbach, F. Caycedo-Soler, S. F. Huelga and M. B. Plenio, *Nat. Phys.*, 2013, **9**, 113–118.
- [6] S. F. Huelga and M. B. Plenio, *Contemporary Physics*, 2013, **54**, 181–207.
- [7] Y. Tanimura and R. Kubo, *J. Phys. Soc. Jpn.*, 1989, **58**, 101–114.
- [8] A. Ishizaki and G. R. Fleming, *J. Chem. Phys.*, 2009, **130**, 234111.

- [9] J. Cao, R. J. Cogdell, D. F. Coker, H.-G. Duan, J. Hauer, U. Kleinekathöfer, T. L. C. Jansen, T. Mančal, R. J. D. Miller, J. P. Ogilvie, V. I. Prokhorenko, T. Renger, H.-S. Tan, R. Tempelaar, M. Thorwart, E. Thyryhaug, S. Westenhoff and D. Zigmantas, *Sci. Adv.*, 2020, **6**, eaaz4888.
- [10] D. Suess, W. T. Strunz and A. Eisfeld, *Journal of Statistical Physics*, 2015, **159**, 1408–1423.
- [11] B. Citty, J. K. Lynd, T. Gera, L. Varvelo and D. I. G. B. Raccach, *The Journal of Chemical Physics*, 2024, **160**, 144105.
- [12] J. Moix, J. Wu, P. Huo, D. Coker and J. Cao, *The Journal of Physical Chemistry Letters*, 2011, **2**, 3045–3052.
- [13] S. Jang, Y.-C. Cheng, D. R. Reichman and J. D. Eaves, *J. Chem. Phys.*, 2008, **129**, 101104.
- [14] H.-G. Duan, V. I. Prokhorenko, R. J. Cogdell, K. Ashraf, A. L. Stevens, M. Thorwart and R. J. D. Miller, *Proc. Natl. Acad. Sci. U.S.A.*, 2017, **114**, 8493–8498.
- [15] A. Strathearn, P. Kirton, D. Kilda, J. Keeling and B. W. Lovett, *Nat. Commun.*, 2018, **9**, 3322.
- [16] T. Baumgratz, M. Cramer and M. B. Plenio, *Phys. Rev. Lett.*, 2014, **113**, 140401.